

Soil node PCB design requirements

AmudarIO | Board 1 | Version 1.1 | 4 September 2026 | Pinout C6-P1

Scope: electrical schematic capture and PCB design for the hardware specified below. Deliver design files only. Firmware, command protocols, cloud services, enclosure design, manufacturing, assembly, commissioning and physical product testing are outside this assignment. Board outline, mounting coordinates and external interface requirements are supplied by the customer.

1 Hardware composition

Block	Required implementation
MCU	ESP32-C6-MINI-1-H4, 4 MB flash; pin map C6-P1
Radio	SX1262-based DX-LR30-900M22S, 3.3 V; external LoRa antenna via SMA
Battery	Removable protected 1S 18650 Li-ion cell, 3.6/3.7 V nominal, 4.2 V charge
Holder	MPD BK-18650-PC2 with retainer; PCB includes holder, not the cell
Solar charger	LTC4079; hardware NTC contacting the battery sidewall
System 3.3 V	TPS63900 buck-boost, always on when SYSTEM is ON
Probe 5 V	TPS61023 with EN and true output disconnect
RS-485	THVD1406 automatic direction; 3.3 V rail switched by TPS22917
Battery ADC	Internal ADC; divider switched on its high side by TPS22917
Service	UART0 header, DTR/RTS circuit, BOOT, RESET and SERVICE WAKE

Do not fit GPS, an external RTC, ADS1115, a fuel gauge, continuously lit LEDs or additional radio interfaces. LiFePO4 is not an interchangeable cell option.

2 Power architecture and charger

A nominal 6 V panel feeds the charger. The protected battery feeds both the 3.3 V converter and 5 V boost through the mechanical SYSTEM switch. SYSTEM OFF disconnects all system loads; the panel, charger, battery and NTC remain connected. Prevent back-power through service and probe connectors.

Configure LTC4079 for 4.2 V and input-voltage regulation based on the panel Vmp. This is fixed-Vmp linear charging, not dynamic MPPT. Set charge current no higher than 250 mA or the panel, cell and thermal limits. Provide hardware charge inhibition at the approved cell temperature limits and on NTC faults. Cell protection must independently cover overvoltage, undervoltage, overload and short circuit; provide reverse-insertion protection.

Calculate converter peak-current margin, charger dissipation and rail capacitance from the customer-supplied loads and panel data. Any charger substitution requires customer approval; do not change the circuit merely to provide a different MPPT method.

3 Complete ESP32 C6 pinout

GPIO is the signal identifier; Pad is the physical ESP32-C6-MINI-1-H4 module pad, not a bare-chip pin or DevKit header. A = ADC input; I = input; O = output. Logic level: 3.3 V.

GPIO	Pad	I/O	Net	Function
0	12	A	VBAT_ADC	Battery voltage, ADC1_CH0
1	13	O	VBAT_DIV_EN	Battery divider high-side switch enable
2	5	O	SENSOR_DOMAIN_EN	Shared enable for 5 V boost and RS-485 power
3	6	O	LORA_RXEN	LoRa RF switch receive control
4	9	-	RESERVED_STRAP_MTMS	Reserved SDIO strap / MTMS
5	10	-	RESERVED_STRAP_MTDI	Reserved SDIO strap / MTDI
6	15	I	SERVICE_WAKE_N	Wake button, connects to GND
7	16	O	LORA_TXEN	LoRa RF switch transmit control
8	22	-	RESERVED_BOOT_STRAP	Reserved boot strap
9	23	I	BOOT_N	BOOT button, programming and recovery only
12	17	O	RS485_TX	UART1 TX to automatic-direction RS-485
13	18	I	RS485_RX	UART1 RX from automatic-direction RS-485
14	19	O	LORA_RESET_N	LoRa reset, active LOW
15	20	-	RESERVED_JTAG_STRAP	Reserved JTAG source-selection strap
16	31	O	UART0_TX	UART0 TX, logs and external USB-to-TTL
17	30	I	UART0_RX	UART0 RX, programming and Serial commands
18	24	O	LORA_SCK	LoRa SPI clock
19	25	I	LORA_MISO	SPI data from LoRa to MCU
20	26	O	LORA_MOSI	SPI data from MCU to LoRa
21	27	O	LORA_NSS	LoRa SPI chip select, active LOW
22	28	I	LORA_DIO1	LoRa interrupt
23	29	I	LORA_BUSY	LoRa busy status

Supply: pad 3 = 3.3 V; pad 8 = EN/CHIP_PU. GND: pads 1, 2, 11, 14, 36-53. NC: pads 4, 7, 21, 32-35.

GPIO10/11 are not exposed. No spare GPIOs remain. Add EN pull-up, RC and supervisor for slow/interrupted supply ramps.

GPIO1 drives the divider-switch enable; GPIO2 drives both the 5 V boost EN and RS-485 power-switch enable. These GPIOs do not carry load current. Do not connect GPIOs directly to battery voltage, 5 V or RS-485 A/B.

4 Pin restrictions and electrical defaults

- GPIO4/5/8/9/15 are strapping pins. Do not load GPIO4/5. Pull GPIO8/9 HIGH through 10 kOhm and GPIO15 LOW through 10 kOhm. BOOT pulls GPIO9 to GND; preserve GPIO8=1 and GPIO9=0 for UART download, GPIO9=1 for normal boot.
- GPIO6/7 reuse pad-JTAG functions. Do not connect an external JTAG interface to GPIO4-7 in this design. Respect the factory eFuse/JTAG source configuration; no eFuse changes are part of PCB design.
- GPIO12/13 default to USB D-/D+. Both UART connections to the switched RS-485 domain require powered-off isolation with specified I_{off}, including during MCU startup. A series resistor alone does not establish isolation.
- GPIO16/17 are dedicated UART0. GPIO18-23 can have startup SDIO pulls. Keep LoRa NSS inactive during startup; do not attach power enables to UART0.

Nets	External reset defaults	Required low-power levels
GPIO1/2	PD about 100 kOhm; LOW	LOW; divider and probe domain OFF
GPIO3/7 RXEN/TXEN	PD about 100 kOhm; LOW	LOW
GPIO14/21 RESET/NSS	PU 47-100 kOhm; HIGH	HIGH with radio in shutdown
GPIO18/20 SCK/MOSI	Keep NSS HIGH	LOW
GPIO6 SERVICE_WAKE_N	PU about 100 kOhm; button to GND	HIGH when released
ADC and UART inputs	Voltage limits and no back-power	No floating or powered-off leakage paths

The pull network must keep external power domains safe while MCU pins are high impedance. Pull values are starting values; check logic margin and leakage. Low-power states are electrical interface conditions, not a software work item.

5 Radio wiring

Signal	DX-LR30 pin	ESP32 GPIO
RXEN	6	3
TXEN	7	7
DIO1	13	22
BUSY	14	23
RESET	15	14
MISO	16	19
MOSI	17	20
SCK	18	18
NSS	19	21

DX-LR30 supply: 3.3 V, pin 9 in the project reference. Leave DIO2/DIO3 unconnected. Verify the purchased module marking and revision drawing before assigning its footprint. LoRa module pin numbers are not ESP32 GPIO numbers.

6 Switched probe rail and battery ADC

GPIO2 HIGH enables TPS61023 and the RS-485 supply switch together; LOW disables both. Add output discharge and probe short-circuit protection. A/B and both UART paths must not back-power disabled rails. Use a keyed probe connector: 1 = +5V_SOIL_SW, 2 = GND/reference, 3 = RS485_A, 4 = RS485_B. Panel connector: PV+ and PV-.

RS-485 direction is automatic; no DE/RE GPIO is allocated. Provide connector-side SM712-class TVS, surge-rated series-resistor footprints and optional 120 ohm termination/bias. Bias belongs to the switched domain. Select termination and AutoDirection timing provisions for the supplied cable and baud rate, including 4800/9600 baud requirements.

Battery divider starting values: high-side switch, 499 + 499 kOhm upper leg, 249 kOhm lower leg and 100 nF ADC-to-GND. GPIO1 controls the switch; GPIO0 is ADC1_CH0. Check divider ratio, RC settling, resistor tolerances, leakage and ADC voltage limits. Include a voltage test pad; no calibration code is required.

7 Service connector

Pin	Board net	Connection
1	GND	Common ground
2	3V3_REF	High-impedance adapter reference only; not a power input
3	UART0_TX / GPIO16	External adapter RX
4	UART0_RX / GPIO17	External adapter TX
5	DTR	Input to onboard two-transistor auto-reset circuit
6	RTS	Input to onboard two-transistor auto-reset circuit

External USB-to-TTL interface: 3.3 V logic only. The board uses its own power source. Do not connect adapter 3.3 V, 5 V or VBUS power outputs to the board. Route DTR/RTS through the standard Espressif two-transistor circuit, not directly to the MCU. Provide independent BOOT (GPIO9) and RESET (EN, module pad 8) buttons or pads. No onboard USB-UART bridge is required.

8 PCB layout and design outputs

Use two-layer FR-4 with a nearly continuous GND plane. Four layers require routing/EMC justification. Route ANT-to-SMA as a short 50 ohm transmission line using fabricator-confirmed geometry, not a series 50 ohm resistor. Keep switching loops/inductors away from ADC/RF, respect module antenna keepouts and place protection beside connectors.

Low-power design limits: complete board ≤ 25 uA with MCU in timer deep sleep, radio in shutdown, probe/RS-485/divider OFF, and charger input absent; inactive states < 1 mA. Active loads and charging are excluded. Budget maximum leakage including charger, protector, supervisor, switches and pulls. Include a removable current-measurement link. Physical current testing is performed outside this PCB-design assignment.

Deliver the native schematic/PCB project with libraries; schematic PDF; BOM with exact manufacturer part numbers and DNP variants; Gerber and NC drill files; fabrication notes and stackup; pick-and-place and assembly drawings; PCB-only STEP model; ERC/DRC results with justified exceptions; RF impedance geometry and electrical sizing notes. Assembly files are design outputs, not an obligation to manufacture or assemble boards.

Customer inputs before design release: exact cell/NTC and panel ratings; probe supply/current/cable data; radio revision and RF band; board outline, holes and connector locations; temperature and surge requirements. Do not invent missing load ratings. Use current manufacturer footprints and electrical design guidance.